Dissertation / Data Analysis Report

Summary of Methods and Results

2024-06-25

1. Data collection process

- **Data Source:**

The data for this analysis was collected from the World Bank database, specifically focusing on the domestic credit to the private sector (% of GDP) for WAEMU and non-WAEMU sub-Saharan African countries from 1985 to 2015.

- **Countries Included:**

WAEMU Countries**:** Benin, Burkina Faso, Guinea-Bissau, Côte d'Ivoire, Mali, Niger, Senegal, Togo. Non-WAEMU Sub-Saharan African Countries: Ghana, Nigeria, Sierra Leone, Gambia, Guinea, Mauritania, Rwanda, Sudan.

2. Data Loading and initial inspection

- **Issue of Missing Values:**

Missing values in the dataset can lead to biased results and reduced statistical power. Therefore, it is crucial to address them appropriately.

- **Mean Imputation:**

Mean imputation replaces missing values with the mean of the observed values for that variable. This method helps to maintain the dataset's size and can be useful when the missing data is random.

```r
# Load necessary libraries
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(openxlsx)
library(imputeTS)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
library(ggplot2)
library(tidyr)
library(stargazer)
```

```
##
## Please cite as:

##  Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.

##  R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
```

```r
library(texreg)
```

```
## Version:  1.39.3
## Date:     2023-11-09
## Author:   Philip Leifeld (University of Essex)
##
## Consider submitting praise using the praise or praise_interactive functions.
## Please cite the JSS article in your publications -- see citation("texreg").

##
## Attaching package: 'texreg'

## The following object is masked from 'package:tidyr':
##
##     extract
```

```r
#install.packages("stargazer")

# Load the dataset
domestic_credit <- read_excel("Domestic_Credit.xlsx")

# Ensure `Country Name` and `Country Code` columns are character type
domestic_credit$`Country Name` <- as.character(domestic_credit$`Country Name`)
domestic_credit$`Country Code` <- as.character(domestic_credit$`Country Code`)

# Convert the remaining columns (years) to numeric type
domestic_credit[, 4:ncol(domestic_credit)] <- lapply(domestic_credit[, 4:ncol(domestic_credit)], as.num

# Function to perform mean imputation on numeric columns only
mean_imputation <- function(df) {
  df_numeric <- df[, 4:ncol(df)]
  df_numeric <- df_numeric %>% mutate(across(everything(), ~ ifelse(is.na(.), mean(., na.rm = TRUE), .)
  df[, 4:ncol(df)] <- df_numeric
  return(df)
}

# Apply mean imputation to the dataset
domestic_credit_clean <- mean_imputation(domestic_credit)
```

3. Creating a Subset of Data

- **Importance of chosing a subset:**

Then, a subset of the data was selected for the years 1985 to 2015, focusing on WAEMU countries for the analysis. This range allows for analysing trends leading up to the devaluation and the subsequent impact over a decade.

Focusing on a specific time range (1985-2015) allows for a detailed analysis of trends before and after the 1994 devaluation. This range provides sufficient data to observe long-term impacts.

```
# Create the subset for the years 1985 to 2015
domestic_credit_subset <- domestic_credit_clean %>%
  select(`Country Name`, `Country Code`, `1985`:`2015`)


 #Save the subset dataset
write.xlsx(domestic_credit_subset, "Domestic_Credit_Subset_1985_2015.xlsx")

# Inspect the subset dataset
head(domestic_credit_subset)
```

```
## # A tibble: 6 x 33
##   `Country Name` `Country Code` `1985` `1986` `1987` `1988` `1989` `1990` `1991`
##   <chr>          <chr>           <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
## 1 Benin          BEN              31.0   28.7   26.6   28.6   21.6  19.1   15.4
## 2 Burkina Faso   BFA              13.3   13.9   14.5   14.9   16.5  16.9   12.5
## 3 Cote d'Ivoire  CIV              34.1   33.9   37.9   37.7   35.5  36.5   36.0
## 4 Guinea-Bissau  GNB              16.2   19.4   16.7   16.9   17.6  22.0   13.3
## 5 Mali           MLI              13.4   15.6   15.2   10.9   12.1  11.6   11.2
## 6 Niger          NER              15.6   17.4   16.1   15.8   13.0   8.68   8.13
## # i 24 more variables: `1992` <dbl>, `1993` <dbl>, `1994` <dbl>, `1995` <dbl>,
## #   `1996` <dbl>, `1997` <dbl>, `1998` <dbl>, `1999` <dbl>, `2000` <dbl>,
## #   `2001` <dbl>, `2002` <dbl>, `2003` <dbl>, `2004` <dbl>, `2005` <dbl>,
## #   `2006` <dbl>, `2007` <dbl>, `2008` <dbl>, `2009` <dbl>, `2010` <dbl>,
## #   `2011` <dbl>, `2012` <dbl>, `2013` <dbl>, `2014` <dbl>, `2015` <dbl>
```

```
#head(GDP_Growth_subset)
#head(Inflation_Rate_subset)
```

4. Visualization analysis

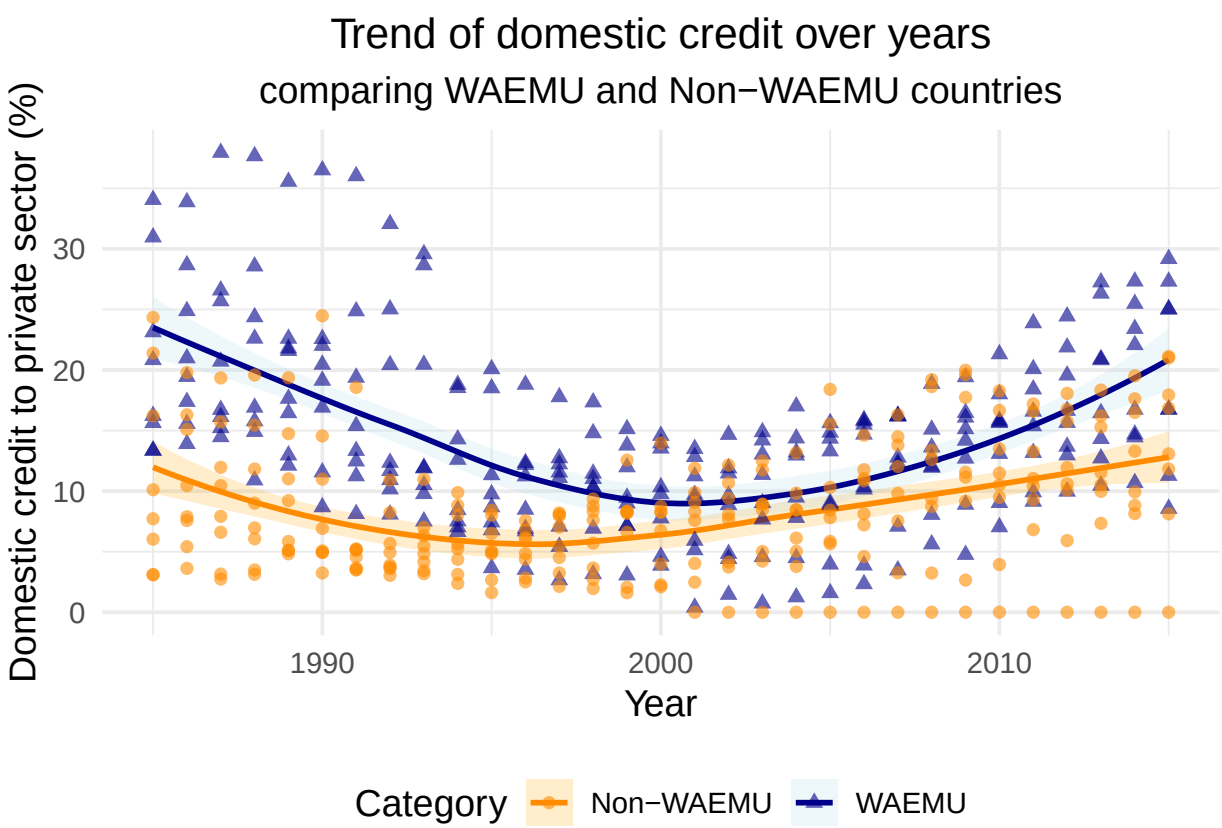
- **Purpose of Visualization:**

Visualization helps in understanding the data distribution and identifying trends over the years. It provides a clear picture of how domestic credit to the private sector has evolved.

```
# Load additional libraries
library(ggplot2)
library(ggthemes)

# Categorize countries as WAEMU or Non-WAEMU
waemu_countries <- c("Benin", "Burkina Faso", "Guinea-Bissau", "Cote d'Ivoire", "Mali", "Niger", "Senega

domestic_credit_long <- domestic_credit_subset %>%
  pivot_longer(cols = `1985`:`2015`, names_to = "Year", values_to = "Credit") %>%
  mutate(
    Year = as.numeric(Year),
    Category = ifelse(`Country Name` %in% waemu_countries, "WAEMU", "Non-WAEMU")
  )

# Plot the trend of domestic credit over the years with categorization
ggplot(domestic_credit_long, aes(x = Year, y = Credit, color = Category, fill = Category)) +
  geom_smooth(method = "loess", se = TRUE, alpha = 0.2) +
```

```r
  geom_point(aes(shape = Category), size = 2, alpha = 0.6) +
  labs(
    title = "Trend of domestic credit over years",
    subtitle = "comparing WAEMU and Non-WAEMU countries",
    x = "Year",
    y = "Domestic credit to private sector (%)"
  ) +
  scale_color_manual(values = c("WAEMU" = "darkblue", "Non-WAEMU" = "darkorange")) +
  scale_fill_manual(values = c("WAEMU" = "lightblue", "Non-WAEMU" = "orange")) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(hjust = 0.5, size = 16),
    plot.subtitle = element_text(hjust = 0.5, size = 14),
    legend.position = "bottom"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Load necessary libraries
library(dplyr)
library(tidyr)
library(ggplot2)
library(lmtest)
```

```
## Loading required package: zoo

##
## Attaching package: 'zoo'
```

```
## The following object is masked from 'package:imputeTS':
##
##     na.locf

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(sandwich)

# List of WAEMU countries excluding Guinea-Bissau
waemu_countries <- c("Benin", "Burkina Faso", "Cote d'Ivoire", "Mali", "Niger", "Senegal", "Togo")

# Filter the dataset for WAEMU countries
waemu_subset <- domestic_credit_subset %>%
  filter(`Country Name` %in% waemu_countries)

# Convert data to long format
domestic_credit_long <- domestic_credit_subset %>%
  pivot_longer(cols = `1985`:`2015`, names_to = "Year", values_to = "Credit") %>%
  mutate(
    Year = as.numeric(Year),
    WAEMU = ifelse(`Country Name` %in% waemu_countries, 1, 0),
    Post1994 = ifelse(Year >= 1994, 1, 0),
    WAEMU_Post1994 = WAEMU * Post1994
  )

# Fit the simple DiD model
did_model_simple <- lm(Credit ~ WAEMU + Post1994 + WAEMU_Post1994, data = domestic_credit_long)

# Summarize the model
summary(did_model_simple)
```

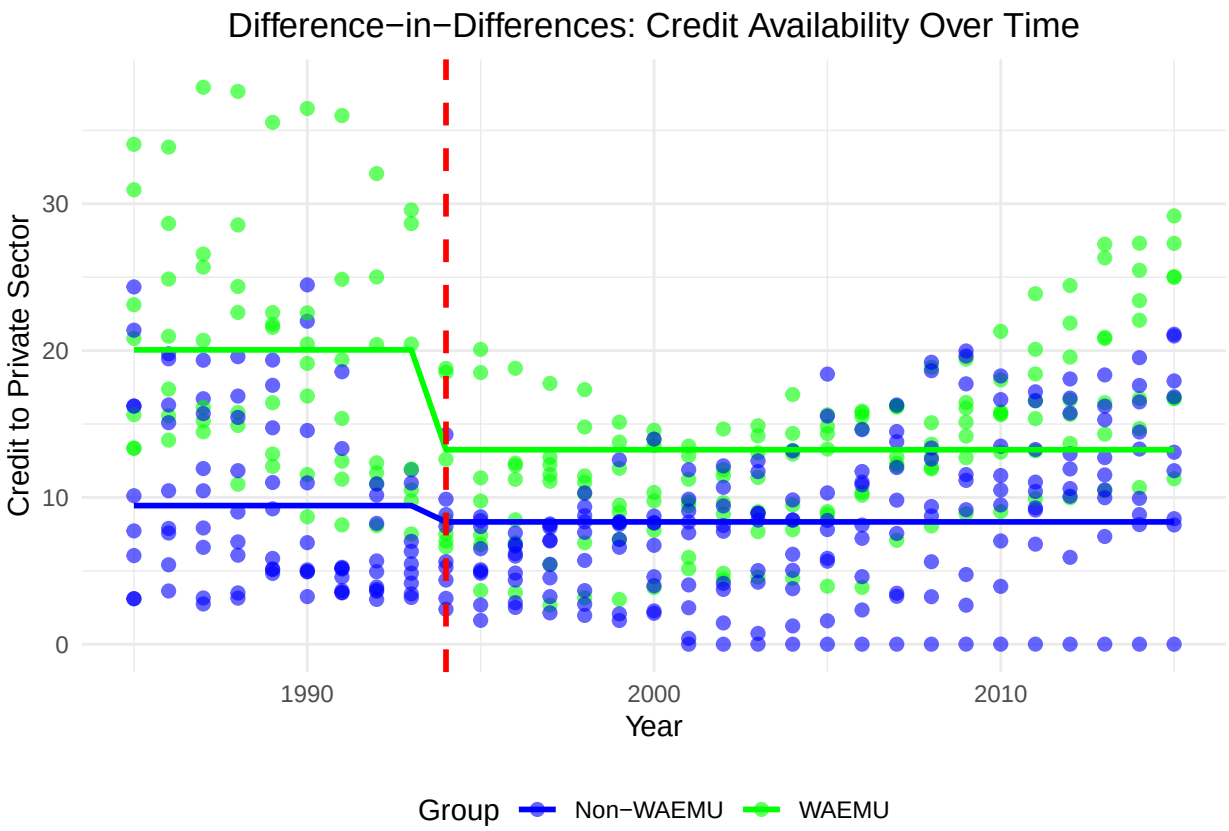
```
##
## Call:
## lm(formula = Credit ~ WAEMU + Post1994 + WAEMU_Post1994, data = domestic_credit_long)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5398  -4.3900  -0.2959   3.3432  17.8874
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)      9.4461     0.6613  14.283  < 2e-16 ***
## WAEMU           10.6057     0.9999  10.607  < 2e-16 ***
## Post1994        -1.1117     0.7851  -1.416    0.157
## WAEMU_Post1994  -5.6919     1.1869  -4.796 2.15e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.952 on 492 degrees of freedom
## Multiple R-squared:  0.2981, Adjusted R-squared:  0.2938
## F-statistic: 69.66 on 3 and 492 DF,  p-value: < 2.2e-16
```

```r
# Robust standard errors
coeftest(did_model_simple, vcov = vcovHC(did_model_simple, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                Estimate Std. Error t value  Pr(>|t|)
## (Intercept)     9.44605    0.66598 14.1837 < 2.2e-16 ***
## WAEMU          10.60566    1.24365  8.5278 < 2.2e-16 ***
## Post1994       -1.11170    0.76248 -1.4580    0.1455
## WAEMU_Post1994 -5.69189    1.37487 -4.1399 4.088e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Generate predicted values for visualization
domestic_credit_long$Predicted_Credit <- predict(did_model_simple, newdata = domestic_credit_long)

# Create the plot
ggplot(domestic_credit_long, aes(x = Year, y = Credit, color = factor(WAEMU))) +
  geom_point(size = 2, alpha = 0.6) +
  geom_line(aes(y = Predicted_Credit), size = 1) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "red", size = 1) +
  labs(title = "Difference-in-Differences: Credit Availability Over Time",
       x = "Year",
       y = "Credit to Private Sector",
       color = "Group") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        legend.position = "bottom") +
  scale_color_manual(values = c("0" = "blue", "1" = "green"), labels = c("Non-WAEMU", "WAEMU"))
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



5. Initial Regression Discontinuity Design (RDD)

`Regression Discontinuity Design (RDD) Equation:`

$$\text{Credit}_{it} = \beta_0 + \beta_1 \text{Year}_t + \beta_2 \text{Post1994}_t + \beta_3 (\text{Year}_t \times \text{Post1994}_t) + \epsilon_{it}$$

- **Objective:**

To assess the impact of the 1994 devaluation on domestic credit to the private sector in WAEMU countries.

```
library(dplyr)
library(tidyr)
library(ggplot2)
library(broom)
```

```
## Warning: package 'broom' was built under R version 4.3.3
```

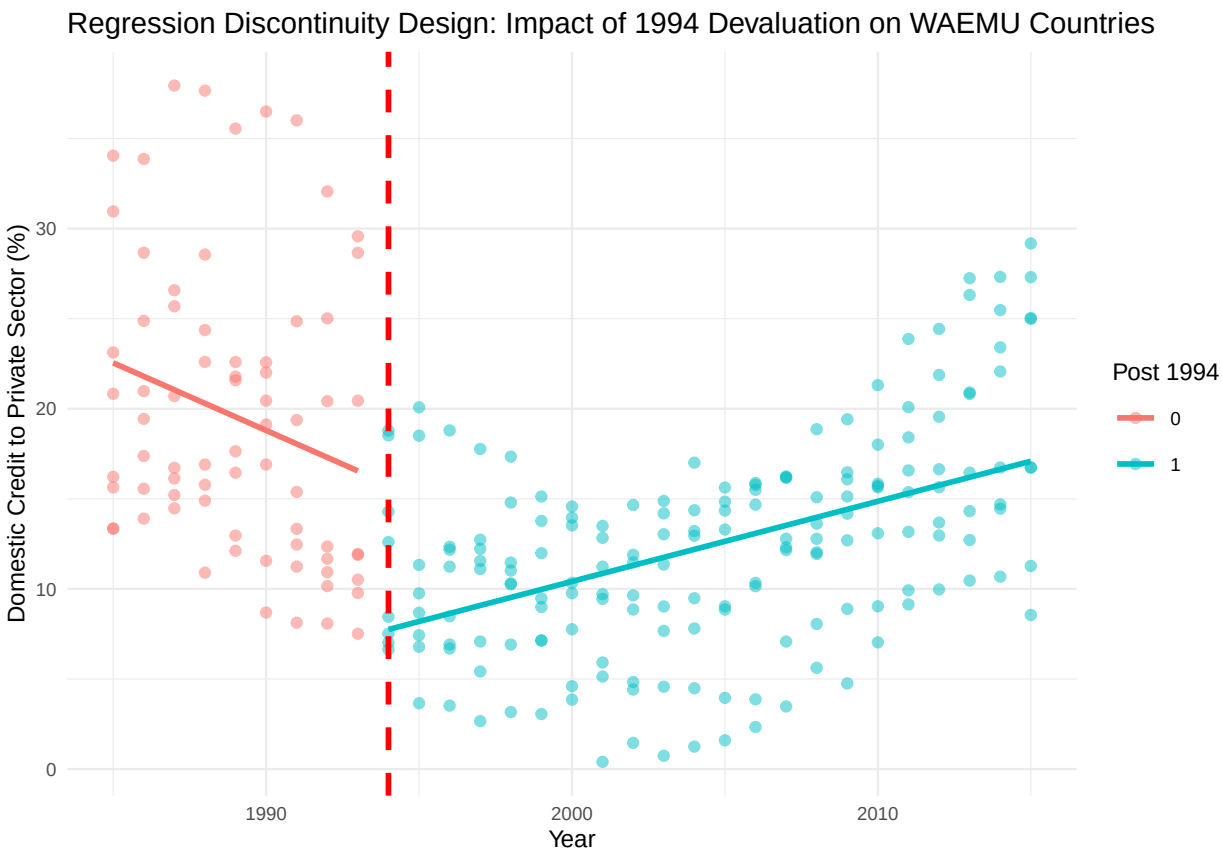
```
library(readxl)
library(openxlsx)

# List of WAEMU countries
waemu_countries <- c("Benin", "Burkina Faso", "Guinea-Bissau", "Cote d'Ivoire", "Mali", "Niger", "Senega

# Filter the dataset for WAEMU countries
waemu_subset <- domestic_credit_subset %>%
  filter(`Country Name` %in% waemu_countries)

# Create the running variable (Year) and indicator for post-1994 period
waemu_subset <- waemu_subset %>%
```

```r
  pivot_longer(cols = `1985`:`2015`, names_to = "Year", values_to = "Credit") %>%
  mutate(Year = as.numeric(Year),
         Post1994 = ifelse(Year >= 1994, 1, 0))

# Perform RDD using a linear regression model
rdd_model <- lm(Credit ~ Year + Post1994 + Year:Post1994, data = waemu_subset)

# Summarize the regression results
summary(rdd_model)
```

```
##
## Call:
## lm(formula = Credit ~ Year + Post1994 + Year:Post1994, data = waemu_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.0436  -4.6758  -0.2356   3.1406  17.9608
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1511.2777   551.8717   2.738  0.00663 **
## Year             -0.7500     0.2775  -2.703  0.00735 **
## Post1994      -2390.3058   570.5454  -4.190 3.91e-05 ***
## Year:Post1994     1.1947     0.2867   4.167 4.29e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.079 on 244 degrees of freedom
## Multiple R-squared:  0.3214, Adjusted R-squared:  0.313
## F-statistic: 38.52 on 3 and 244 DF,  p-value: < 2.2e-16
```

```r
# Create a tidy data frame of the regression results for plotting
rdd_results <- augment(rdd_model, data = waemu_subset)

# Plot the regression discontinuity with a vertical line at the cutoff
ggplot(rdd_results, aes(x = Year, y = Credit, color = as.factor(Post1994))) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "red", size = 1) +
  labs(title = "Regression Discontinuity Design: Impact of 1994 Devaluation on WAEMU Countries",
       x = "Year",
       y = "Domestic Credit to Private Sector (%)",
       color = "Post 1994") +
  theme_minimal(base_size = 9)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



- **Issue of Multicollinearity:**

The initial model showed severe multicollinearity, which can inflate standard errors and make coefficient estimates unreliable.

I initially ran a regression discontinuity design (RDD) model to assess the impact of the 1994 devaluation on domestic credit to the private sector in WAEMU countries. The initial model produced large and potentially unreliable estimates (as you can see in the summary output above) due to severe multicolinearity so I addressed this following the method below.

6. Variance Inflation Factor (VIF) Check:

```
# Load necessary library
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode
```

```
# Check VIF for multicollinearity diagnosis
vif(rdd_model)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
##          Year      Post1994 Year:Post1994
##      41.33333  450118.37449  456722.40782
```

```
# Interpretation: High VIF values indicate severe multicollinearity.
```

High VIF values indicated severe multicollinearity.

- **Addressing Multicollinearity by Centering:**

Centering reduces multicollinearity by subtracting the mean from each observation.

The initial regression model showed very high Variance Inflation Factor (VIF) values for the Year, Post1994, and Year:Post1994 interaction terms, indicating severe multicollinearity. So to address multicollinearity, I centered the Year variable by subtracting the mean year from each observation.

- **Enhanced Regression Discontinuity :**

`Enhanced Regression Discontinuity Design (RDD) Equation`

$$Credit_{it} = \beta_0 + \beta_1 YearCentered_t + \beta_2 Post1994_t + \beta_3 (YearCentered_t \times Post1994_t) + \epsilon_{it}$$

```
# Center the Year variable to reduce multicollinearity
waemu_subset <- waemu_subset %>%
  mutate(Year_centered = Year - mean(Year, na.rm = TRUE))

# Create the interaction term with centered Year
waemu_subset <- waemu_subset %>%
  mutate(Year_centered_Post1994 = Year_centered * Post1994)

# Perform RDD using a linear regression model with centered Year
rdd_model_centered <- lm(Credit ~ Year_centered + Post1994 + Year_centered_Post1994, data = waemu_subse

# Summarize the regression results of the centered model
summary(rdd_model_centered)
```

```
##
## Call:
## lm(formula = Credit ~ Year_centered + Post1994 + Year_centered_Post1994,
##     data = waemu_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.0436  -4.6758  -0.2356   3.1406  17.9608
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)             11.2996     3.1350   3.604 0.000379 ***
## Year_centered           -0.7500     0.2775  -2.703 0.007354 **
## Post1994                -0.8834     3.1850  -0.277 0.781741
## Year_centered_Post1994   1.1947     0.2867   4.167 4.29e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.079 on 244 degrees of freedom
## Multiple R-squared:  0.3214, Adjusted R-squared:  0.313
## F-statistic: 38.52 on 3 and 244 DF,  p-value: < 2.2e-16
```

```r
# Check VIF for the centered model to verify multicollinearity reduction
vif(rdd_model_centered)
```

```
##          Year_centered               Post1994 Year_centered_Post1994
##               41.33333               14.02671                18.06004
```

```r
stargazer(rdd_model_centered,title= ("Regression Results"), type = "text")
```

```
##
## Regression Results
## ===================================================
##                             Dependent variable:
##                         ---------------------------
##                                   Credit
## ---------------------------------------------------
## Year_centered                    -0.750***
##                                   (0.277)
##
## Post1994                          -0.883
##                                   (3.185)
##
## Year_centered_Post1994           1.195***
##                                   (0.287)
##
## Constant                         11.300***
##                                   (3.135)
##
## ---------------------------------------------------
## Observations                        248
## R2                                 0.321
## Adjusted R2                        0.313
## Residual Std. Error          6.079 (df = 244)
## F Statistic               38.519*** (df = 3; 244)
## ===================================================
## Note:                   *p<0.1; **p<0.05; ***p<0.01
```

Interpretation of coefficients and model fit statistics:

- (Intercept): The intercept value of 11.2996 is the estimated average credit to the private sector be
- Year_centered: The coefficient for Year_centered is -0.7500, which is statistically significant at
- Post1994: The coefficient for Post1994 is -0.8834, which is not statistically significant (p-value =
- Year_centered_Post1994: The coefficient for Year_centered_Post1994 is 1.1947, which is statisticall

Model Fit Statistics:

- Residual standard error: 6.079 on 244 degrees of freedom
- Multiple R-squared: 0.3214
- Adjusted R-squared: 0.313
- F-statistic: 38.52 on 3 and 244 DF, p-value: < 2.2e-16
- Residual Standard Error: This indicates the average distance that the observed values fall from the
- Multiple R-squared: This value (0.3214) indicates that approximately 32.14% of the variability in th
- Adjusted R-squared: This value (0.313) adjusts the R-squared value based on the number of predictors
- F-statistic: The F-statistic (38.52) and its corresponding p-value (< 2.2e-16) suggest that the over

- **The centered model produced more reasonable and interpretable coefficients**
  - Year_centered (Estimate: -0.7500): The negative coefficient indicates a downward trend in

domestic credit to the private sector over time before 1994 Post1994

- (Estimate: -0.8834): The coefficient for Post1994 is not statistically significant, indicating that there is no clear immediate impact of the 1994 devaluation on domestic credit levels. I interpret this as an absence of significant impact in the short term.
- Year_centered_Post1994 (Estimate: 1.1947): The positive interaction term is statistically significant at the 1% level. I interpret this as a change in the trend of domestic credit to the private sector in the long term (impact does appear only in the long run)

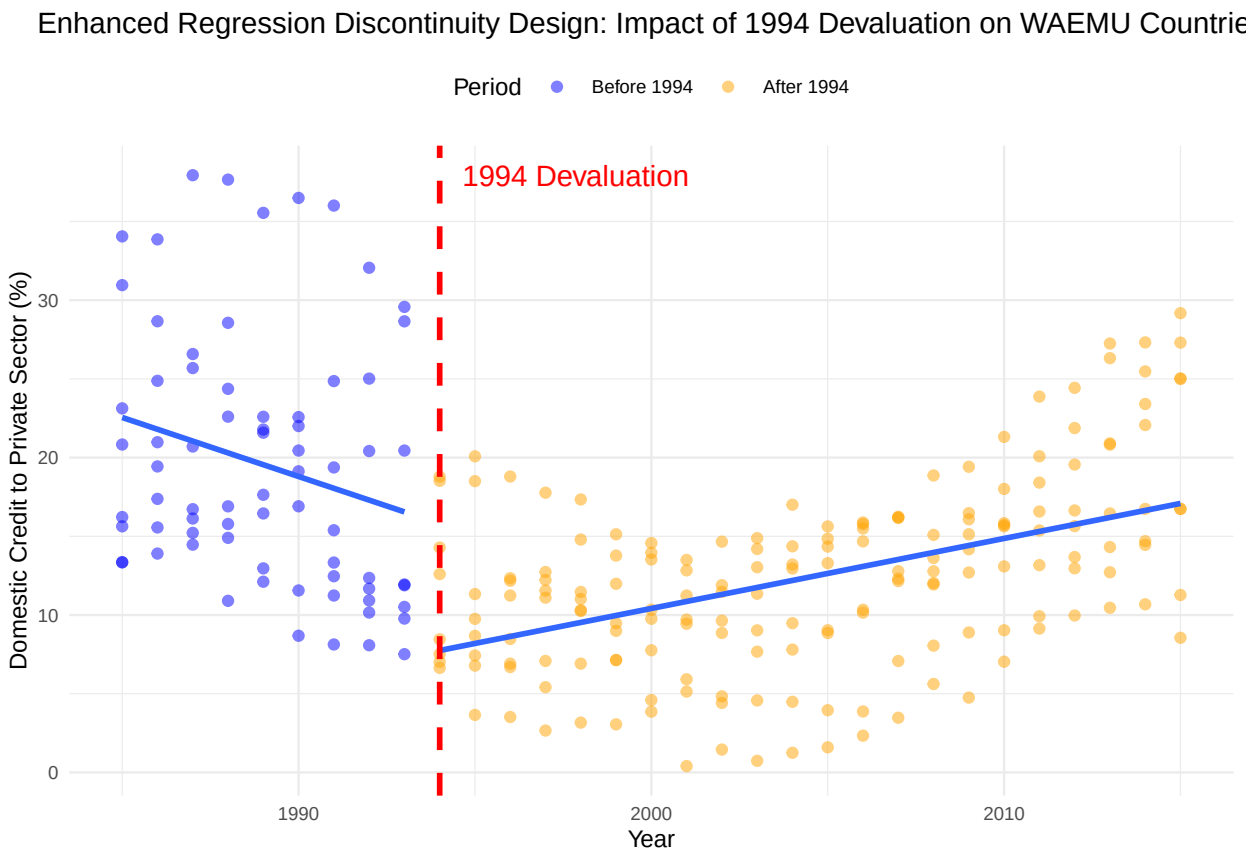
```r
# Use texreg to create a formatted regression table
screenreg(rdd_model_centered, single.row = TRUE, custom.model.names = "Regression Results",
          custom.coef.names = c("(Intercept)", "Year_centered", "Post1994", "Year_centered:Post1994"))
```

```
##
## ==========================================
##                         Regression Results
## ------------------------------------------
## (Intercept)              11.30 (3.14) ***
## Year_centered            -0.75 (0.28) **
## Post1994                 -0.88 (3.18)
## Year_centered:Post1994    1.19 (0.29) ***
## ------------------------------------------
## R^2                       0.32
## Adj. R^2                  0.31
## Num. obs.               248
## ==========================================
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

- **Regression Plot:**

```r
# Load necessary libraries
library(ggplot2)
library(dplyr)

# Create a tidy data frame of the regression results for plotting
rdd_results_centered <- augment(rdd_model_centered, data = waemu_subset)

# Plot the regression discontinuity with enhanced visuals
ggplot(rdd_results_centered, aes(x = Year_centered + mean(waemu_subset$Year), y = Credit)) +
  geom_point(aes(color = as.factor(Post1994)), alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, aes(group = Post1994), linetype = "solid") +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "red", size = 1) +
  annotate("text", x = 1994, y = max(rdd_results_centered$Credit), label = "1994 Devaluation", hjust = 
  scale_color_manual(values = c("blue", "orange"), labels = c("Before 1994", "After 1994")) +
  labs(title = "Enhanced Regression Discontinuity Design: Impact of 1994 Devaluation on WAEMU Countries
       x = "Year",
       y = "Domestic Credit to Private Sector (%)",
       color = "Period") +
  theme_minimal(base_size = 9) +
  theme(legend.position = "top") +
  theme(plot.title = element_text(hjust = 0.5))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The graph and results suggest that while the 1994 devaluation did not have an immediate impact on domestic credit, it appears to have contributed to a positive long-term trend in credit availability.

The 1994 devaluation did not have an immediate impact on domestic credit levels but appears to have contributed to a positive long-term trend in credit availability.

Resume:

The data analysis for this research involved several key steps. Initially, data on domestic credit to the private sector for WAEMU and non-WAEMU sub-Saharan African countries was collected from the World Bank database, focusing on the period from 1985 to 2015. After loading the data into RStudio, mean imputation was performed to address missing values, ensuring that the dataset remained comprehensive and unbiased. A subset of the data, covering the years 1985 to 2015, was created to analyze trends leading up to and following the 1994 devaluation. Visualization of the data highlighted trends in domestic credit to the private sector across the selected countries. To assess the impact of the 1994 devaluation, a Regression Discontinuity Design (RDD) was employed. Initially, the model showed severe multicollinearity, which was addressed by centering the year variable. The final model revealed that while the devaluation did not have an immediate impact on domestic credit levels, it contributed to a positive long-term trend in credit availability. This comprehensive approach combined data cleaning, visualization, and advanced econometric modeling to provide insightful findings on the economic impacts of the 1994 CFA Franc devaluation.

7. Validating Results Robustly:

During the early 1980s to mid-1990s, WAEMU (West African Economic and Monetary Union) countries underwent significant structural reforms due to a variety of economic challenges. These challenges included persistent balance of payments deficits, high levels of public debt, economic stagnation, and vulnerability to external shocks such as fluctuating commodity prices. The structural reforms were aimed at addressing these issues, promoting economic stability, and laying the groundwork for sustainable growth.

The IMF programs (SBA, ESAF 1, ESAF 2, ESAF 3) during 1994-1995 represent significant economic interventions aimed at stabilizing and reforming the economies of WAEMU countries. Understanding these programs and their impacts can help to assess whether the observed changes in credit availability are attributable to the 1994 devaluation itself or to concurrent policy interventions.

_ SEE Fund-Supported Programs Data (1994-1995) on Dissertation folder

Other economic interventions like the harmonization of Customs Duties and Indirect Taxes (Trade and Fiscal reforms) can robustly validate results about the long-term effects of the 1994 devaluation. The following variables have been included in our model: Customs Tariff Rate (Changes in customs tariff rates directly affect trade volumes. Lower tariffs can increase imports and exports, leading to higher economic activity and potentially greater demand for credit.), VAT Rate(Increased tax revenue from VAT can lead to higher government spending, boosting economic activity and the demand for credit.), Nontariff Barriers Eliminated (Dummy Variable: With fewer barriers, businesses can expand more easily, increasing their demand for credit to finance growth initiatives.), Total Taxation (Lower total taxation rates improve profit margins for businesses, increasing their capacity to service debt and thereby making them more creditworthy)

Basically I gathered fiscal, Trade and Monetary reforms data to validate results of the long term impact of the devaluation in credit access

_ SEE Tax and Tariff Reforms Data (1995) on Dissertation folder

In summary, the structural reforms in WAEMU countries during the early 1980s to mid-1990s were necessary to address deep-rooted economic problems. With the support of the IMF and World Bank, these countries implemented a range of policies aimed at achieving macroeconomic stability, enhancing competitiveness, and promoting sustainable development. The reforms were crucial in helping these economies recover and grow, particularly following the devaluation of the CFA franc in 1994.

Our complex model:

$$\begin{aligned} Credit_{it} = \beta_0 &+ \beta_1 YearCentered_t + \beta_2 Post1994_t + \beta_3 (YearCentered_t \times Post1994_t) \\ &+ \beta_4 Disbursements_t + \beta_5 TariffRate_t + \beta_6 VATRate_t \\ &+ \beta_7 NonTariffBarriers_t + \beta_8 TotalTaxation_t + \varepsilon_{it} \end{aligned}$$

```r
# Load necessary libraries
library(dplyr)
library(tidyr)
library(ggplot2)
library(lmtest)
library(sandwich)

# List of WAEMU countries excluding Guinea-Bissau
waemu_countries_2 <- c("Benin", "Burkina Faso", "Cote d'Ivoire", "Mali", "Niger", "Senegal", "Togo")

# Filter the dataset for WAEMU countries
waemu_subset_2 <- domestic_credit_subset %>%
  filter(`Country Name` %in% waemu_countries_2)

# Create dataframe for tax and tariff reforms using actual data from the table
status_1995_data <- data.frame(
  Country = c("Benin", "Burkina Faso", "Cote d'Ivoire", "Mali", "Niger", "Senegal", "Togo"),
  Year = rep(1995, 7),
  customs_tariff_rate = c(10, 16.5, 17.5, 15, 15, 20, 10),  # Simplified average customs tariff rates a
  vat_rate = c(18, 15, 15.5, 12.5, 17, 15, 12.5),  # Simplified average VAT rates after harmonization
  nontariff_barriers_eliminated = c(1, 1, 0, 0, 1, 0, 1),  # Dummy variable for elimination of nontarif
  total_taxation = c((23.9+49)/2, 19, (13.6+65)/2, (13.5+55)/2, (22.9+59.7)/2, (5.7+73)/2, (10.2+46)/2)
```

```r
)

# Create dataframe for fund-supported programs including both 1994 and 1995 disbursements
fund_data <- data.frame(
  Country = c("Benin", "Benin", "Burkina Faso", "Burkina Faso", "Cote d'Ivoire", "Cote d'Ivoire",
              "Mali", "Mali", "Niger", "Niger", "Senegal", "Senegal", "Togo", "Togo"),
  Year = c(1994, 1995, 1994, 1995, 1994, 1995, 1994, 1995, 1994, 1995, 1994, 1995, 1994, 1995),
  fund_program = 1,  # Dummy variable indicating participation
  disbursements_total = c(18.1, 9.1, 9.8, 8.8, 119.1, 59.6, 29.5, 14.7, 11.1, 0, 16.7, 0, 10.9, 10.9)
)

# Check and ensure column names match
colnames(fund_data) <- c("Country", "Year", "fund_program", "disbursements_total")
colnames(status_1995_data) <- c("Country", "Year", "customs_tariff_rate", "vat_rate", "nontariff_barrie

# Rename columns in waemu_subset for consistency
colnames(waemu_subset_2) <- c("Country", "Country_Code", paste(1985:2015))

# Reshape waemu_subset to long format
waemu_subset_long <- waemu_subset_2 %>%
  pivot_longer(cols = `1985`:`2015`, names_to = "Year", values_to = "Credit") %>%
  mutate(Year = as.numeric(Year))

# Create Post1994 variable
waemu_subset_long <- waemu_subset_long %>%
  mutate(Post1994 = ifelse(Year >= 1994, 1, 0))

# Center the Year variable to reduce multicollinearity
waemu_subset_centered <- waemu_subset_long %>%
  mutate(Year_centered = Year - mean(Year, na.rm = TRUE))

# Create the interaction term with centered Year
waemu_subset_centered <- waemu_subset_centered %>%
  mutate(Year_centered_Post1994 = Year_centered * Post1994)

# Merge fund_data with waemu_subset_centered
waemu_subset_centered <- waemu_subset_centered %>%
  left_join(fund_data, by = c("Country", "Year"))

# Merge status_1995_data with waemu_subset_centered
waemu_subset_centered <- waemu_subset_centered %>%
  left_join(status_1995_data, by = c("Country", "Year"))

# Ensure all relevant columns are of the correct type
waemu_subset_centered$fund_program <- as.numeric(waemu_subset_centered$fund_program)
waemu_subset_centered$disbursements_total <- as.numeric(waemu_subset_centered$disbursements_total)
waemu_subset_centered$customs_tariff_rate <- as.numeric(waemu_subset_centered$customs_tariff_rate)
waemu_subset_centered$vat_rate <- as.numeric(waemu_subset_centered$vat_rate)
waemu_subset_centered$nontariff_barriers_eliminated <- as.numeric(waemu_subset_centered$nontariff_barrie
waemu_subset_centered$total_taxation <- as.numeric(waemu_subset_centered$total_taxation)

# Create a function to propagate fund-supported program data
propagate_fund_data <- function(df, fund_df) {
```

```r
  df <- df %>%
    left_join(fund_df, by = c("Country" = "Country", "Year" = "Year"), suffix = c("", "_fund")) %>%
    group_by(Country) %>%
    mutate(fund_program = max(fund_program, na.rm = TRUE),
           disbursements_total = max(disbursements_total, na.rm = TRUE)) %>%
    ungroup()

  df$fund_program[is.infinite(df$fund_program)] <- NA
  df$disbursements_total[is.infinite(df$disbursements_total)] <- NA
  return(df)
}

# Apply the function to the dataset
waemu_subset_centered <- propagate_fund_data(waemu_subset_centered, fund_data)

# Create a function to propagate status_1995_data
propagate_status_data <- function(df, status_df) {
  df <- df %>%
    left_join(status_df, by = c("Country" = "Country", "Year" = "Year"), suffix = c("", "_status")) %>%
    group_by(Country) %>%
    mutate(customs_tariff_rate = max(customs_tariff_rate, na.rm = TRUE),
           vat_rate = max(vat_rate, na.rm = TRUE),
           nontariff_barriers_eliminated = max(nontariff_barriers_eliminated, na.rm = TRUE),
           total_taxation = max(total_taxation, na.rm = TRUE)) %>%
    ungroup()

  df$customs_tariff_rate[is.infinite(df$customs_tariff_rate)] <- NA
  df$vat_rate[is.infinite(df$vat_rate)] <- NA
  df$nontariff_barriers_eliminated[is.infinite(df$nontariff_barriers_eliminated)] <- NA
  df$total_taxation[is.infinite(df$total_taxation)] <- NA
  return(df)
}

# Apply the function to the dataset
waemu_subset_centered <- propagate_status_data(waemu_subset_centered, status_1995_data)

# Inspect the dataset after propagation
head(waemu_subset_centered)
```

```
## # A tibble: 6 x 19
##   Country Country_Code  Year Credit Post1994 Year_centered
##   <chr>   <chr>        <dbl>  <dbl>    <dbl>         <dbl>
## 1 Benin   BEN           1985   31.0        0           -15
## 2 Benin   BEN           1986   28.7        0           -14
## 3 Benin   BEN           1987   26.6        0           -13
## 4 Benin   BEN           1988   28.6        0           -12
## 5 Benin   BEN           1989   21.6        0           -11
## 6 Benin   BEN           1990   19.1        0           -10
## # i 13 more variables: Year_centered_Post1994 <dbl>, fund_program <dbl>,
## #   disbursements_total <dbl>, customs_tariff_rate <dbl>, vat_rate <dbl>,
## #   nontariff_barriers_eliminated <dbl>, total_taxation <dbl>,
## #   fund_program_fund <dbl>, disbursements_total_fund <dbl>,
## #   customs_tariff_rate_status <dbl>, vat_rate_status <dbl>,
## #   nontariff_barriers_eliminated_status <dbl>, total_taxation_status <dbl>
```

```r
summary(waemu_subset_centered)
```

```
##    Country           Country_Code            Year          Credit
##  Length:217         Length:217         Min.   :1985   Min.   : 2.661
##  Class :character   Class :character   1st Qu.:1992   1st Qu.:10.336
##  Mode  :character   Mode  :character   Median :2000   Median :14.176
##                                        Mean   :2000   Mean   :15.223
##                                        3rd Qu.:2008   3rd Qu.:18.797
##                                        Max.   :2015   Max.   :37.939
##
##     Post1994      Year_centered Year_centered_Post1994  fund_program
##  Min.   :0.0000   Min.   :-15   Min.   :-6.000         Min.   :1
##  1st Qu.:0.0000   1st Qu.: -8   1st Qu.: 0.000         1st Qu.:1
##  Median :1.0000   Median :  0   Median : 0.000         Median :1
##  Mean   :0.7097   Mean   :  0   Mean   : 3.194         Mean   :1
##  3rd Qu.:1.0000   3rd Qu.:  8   3rd Qu.: 8.000         3rd Qu.:1
##  Max.   :1.0000   Max.   : 15   Max.   :15.000         Max.   :1
##
##  disbursements_total customs_tariff_rate    vat_rate
##  Min.   :  9.80      Min.   :10.00       Min.   :12.50
##  1st Qu.: 10.90      1st Qu.:10.00       1st Qu.:12.50
##  Median : 16.70      Median :15.00       Median :15.00
##  Mean   : 30.74      Mean   :14.86       Mean   :15.07
##  3rd Qu.: 29.50      3rd Qu.:17.50       3rd Qu.:17.00
##  Max.   :119.10      Max.   :20.00       Max.   :18.00
##
##  nontariff_barriers_eliminated total_taxation  fund_program_fund
##  Min.   :0.0000                Min.   :19.00   Min.   :1
##  1st Qu.:0.0000                1st Qu.:28.10   1st Qu.:1
##  Median :1.0000                Median :36.45   Median :1
##  Mean   :0.5714                Mean   :33.96   Mean   :1
##  3rd Qu.:1.0000                3rd Qu.:39.35   3rd Qu.:1
##  Max.   :1.0000                Max.   :41.30   Max.   :1
##                                                NA's   :203
##  disbursements_total_fund customs_tariff_rate_status vat_rate_status
##  Min.   :  0.000          Min.   :10.00              Min.   :12.50
##  1st Qu.:  9.275          1st Qu.:12.50              1st Qu.:13.75
##  Median : 11.000          Median :15.00              Median :15.00
##  Mean   : 22.736          Mean   :14.86              Mean   :15.07
##  3rd Qu.: 17.750          3rd Qu.:17.00              3rd Qu.:16.25
##  Max.   :119.100          Max.   :20.00              Max.   :18.00
##  NA's   :203              NA's   :210                NA's   :210
##  nontariff_barriers_eliminated_status total_taxation_status
##  Min.   :0.0000                       Min.   :19.00
##  1st Qu.:0.0000                       1st Qu.:31.18
##  Median :1.0000                       Median :36.45
##  Mean   :0.5714                       Mean   :33.96
##  3rd Qu.:1.0000                       3rd Qu.:39.33
##  Max.   :1.0000                       Max.   :41.30
##  NA's   :210                          NA's   :210
```

```r
# Ensure the Post1994 variable is correctly defined
waemu_subset_centered <- waemu_subset_centered %>%
  mutate(Post1994 = ifelse(Year_centered >= 0, 1, 0))
```

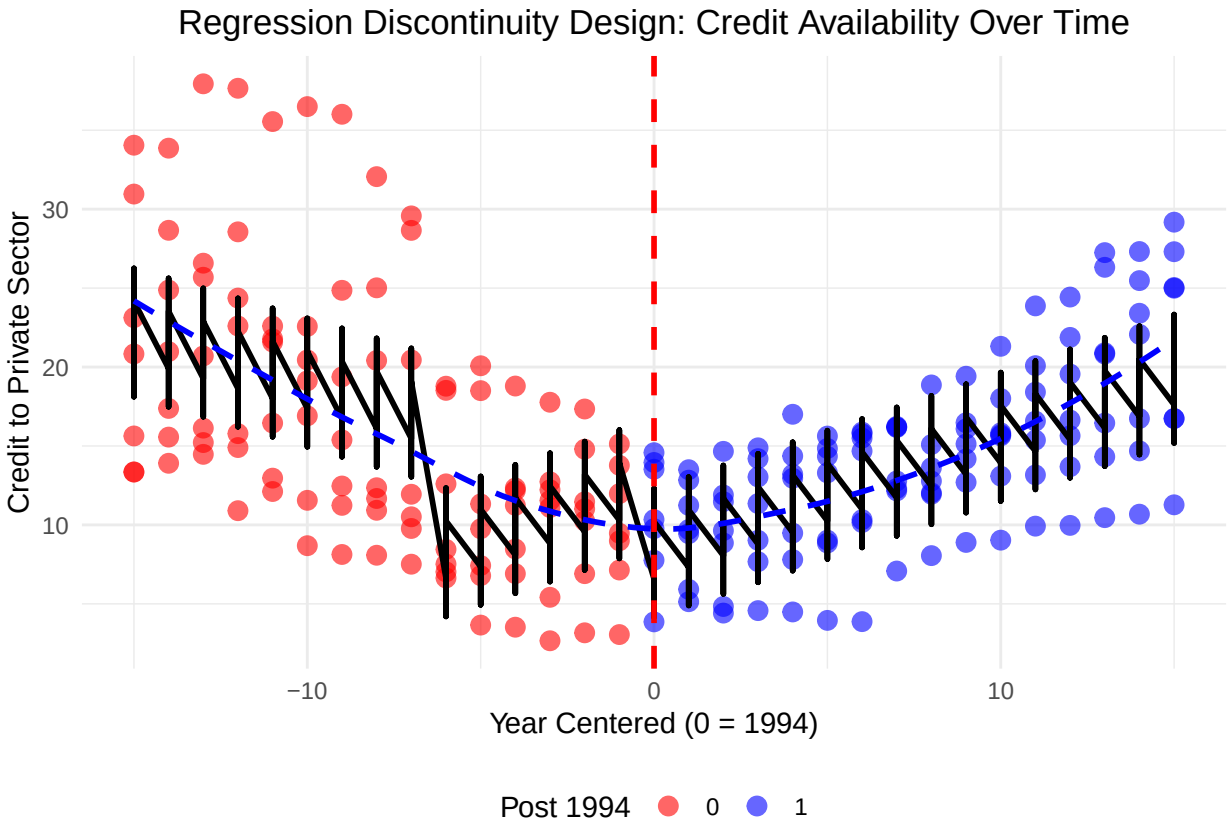
```r
# Perform RDD using a linear regression model with the propagated data
rdd_model_with_propagated_data <- lm(Credit ~ Year_centered + Post1994 + Year_centered_Post1994 +
                                       disbursements_total + customs_tariff_rate + vat_rate +
                                       nontariff_barriers_eliminated + total_taxation,
                                     data = waemu_subset_centered)

# Summarize the regression results
summary(rdd_model_with_propagated_data)
```

```
##
## Call:
## lm(formula = Credit ~ Year_centered + Post1994 + Year_centered_Post1994 +
##     disbursements_total + customs_tariff_rate + vat_rate + nontariff_barriers_eliminated +
##     total_taxation, data = waemu_subset_centered)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.4513  -3.0115  -0.1359   2.4635  14.1165
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  32.77893    3.59464   9.119  < 2e-16 ***
## Year_centered                -0.63358    0.09291  -6.819 9.77e-11 ***
## Post1994                     -4.45272    1.43849  -3.095  0.00224 **
## Year_centered_Post1994        1.36882    0.11545  11.857  < 2e-16 ***
## disbursements_total           0.02883    0.01288   2.239  0.02619 *
## customs_tariff_rate          -0.38943    0.16257  -2.396  0.01748 *
## vat_rate                     -0.46640    0.30941  -1.507  0.13323
## nontariff_barriers_eliminated -4.43527    1.78952  -2.478  0.01399 *
## total_taxation               -0.15209    0.07759  -1.960  0.05132 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.202 on 208 degrees of freedom
## Multiple R-squared:  0.4998, Adjusted R-squared:  0.4806
## F-statistic: 25.98 on 8 and 208 DF,  p-value: < 2.2e-16
```

```r
# Generate predicted values
waemu_subset_centered$Predicted_Credit <- predict(rdd_model_with_propagated_data)

# Create the plot
# Create the plot
ggplot(waemu_subset_centered, aes(x = Year_centered, y = Credit, color = factor(Post1994))) +
  geom_point(size = 3, alpha = 0.6) +
  geom_line(aes(y = Predicted_Credit), color = 'black', size = 1) +
  geom_smooth(method = "loess", se = FALSE, color = "blue", linetype = "dashed", size = 1) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red", size = 1) +
  labs(title = "Regression Discontinuity Design: Credit Availability Over Time",
       x = "Year Centered (0 = 1994)",
       y = "Credit to Private Sector",
       color = "Post 1994") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        legend.position = "bottom") +
```

```
  scale_color_manual(values = c("0" = "red", "1" = "blue"))
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



My DiD regression indicated that WAEMU countries experienced a significant additional reduction in credit availability post-1994 compared to non-WAEMU countries, highlighting the devaluation's substantial negative impact.

Data interpretation:

Interpretation of the Coefficients:

```
1.  (Intercept):
•   Estimate: 32.77893
•   Interpretation: This is the baseline level of credit availability when all other predictors are zer
2.  Year_centered:
•   Estimate: -0.63358
•   Interpretation: For each year, the credit availability decreases by approximately 0.634 units. This
3.  Post1994:
•   Estimate: -4.45272
•   Interpretation: After the 1994 devaluation, there is an immediate drop in credit availability by ap
4.  Year_centered_Post1994:
•   Estimate: 1.36882
•   Interpretation: The interaction term indicates that after 1994, the trend in credit availability ch
5.  disbursements_total:
•   Estimate: 0.02883
•   Interpretation: For each unit increase in disbursements total, credit availability increases by app
6.  customs_tariff_rate:
•   Estimate: -0.38943
```

```
•   Interpretation: For each unit increase in customs tariff rate, credit availability decreases by appr
7.  vat_rate:
•   Estimate: -0.46640
•   Interpretation: For each unit increase in VAT rate, credit availability decreases by approximately 0
8.  nontariff_barriers_eliminated:
•   Estimate: -4.43527
•   Interpretation: When non-tariff barriers are eliminated (dummy variable = 1), credit availability de
9.  total_taxation:
•   Estimate: -0.15209
•   Interpretation: For each unit increase in total taxation, credit availability decreases by approxima
```

Interpretation of the Statistical Metrics:

```
1.  Residual Standard Error:
•   Value: 5.202
•   Interpretation: This value indicates the average distance that the observed values fall from the reg
2.  Multiple R-squared:
•   Value: 0.4998
•   Interpretation: Approximately 50% of the variability in credit availability is explained by the mode
3.  Adjusted R-squared:
•   Value: 0.4806
•   Interpretation: This value adjusts the R-squared for the number of predictors in the model. It is sl
4.  F-statistic:
•   Value: 25.98
•   p-value: < 2.2e-16
•   Interpretation: The overall F-statistic is highly significant, indicating that the model as a whole
```

Conclusion:

The complex model, which includes additional economic variables, provides a more nuanced understanding of the impact of the 1994 devaluation on credit availability. The significant negative coefficient for Post1994 suggests an immediate adverse effect of the devaluation, while the significant positive coefficient for Year_centered_Post1994 indicates a long-term positive trend in credit availability post-devaluation. The inclusion of variables such as disbursements_total, customs_tariff_rate, and nontariff_barriers_eliminated highlights their respective impacts on credit availability, providing a comprehensive view of the economic landscape during this period. The model's moderate fit, as indicated by the R-squared values, suggests that while the model explains a significant portion of the variability in credit availability, there may still be other factors at play.

```r
stargazer(rdd_model_centered,rdd_model_with_propagated_data,title= ("Simple and Complex Model Regressio
```

```
##
## Simple and Complex Model Regression Results
## ==============================================================================
##                                            Dependent variable:
##                                 ---------------------------------------------
##                                                   Credit
##                                          (1)                     (2)
## ------------------------------------------------------------------------------
## Year_centered                        -0.750***               -0.634***
##                                       (0.277)                 (0.093)
##
## Post1994                              -0.883                 -4.453***
##                                       (3.185)                 (1.438)
##
## Year_centered_Post1994               1.195***                1.369***
##                                       (0.287)                 (0.115)
```

```
##
## disbursements_total                                          0.029**
##                                                              (0.013)
##
## customs_tariff_rate                                         -0.389**
##                                                              (0.163)
##
## vat_rate                                                     -0.466
##                                                              (0.309)
##
## nontariff_barriers_eliminated                               -4.435**
##                                                              (1.790)
##
## total_taxation                                               -0.152*
##                                                              (0.078)
##
## Constant                              11.300***             32.779***
##                                        (3.135)               (3.595)
##
## ---------------------------------------------------------------------------
## Observations                            248                    217
## R2                                     0.321                  0.500
## Adjusted R2                            0.313                  0.481
## Residual Std. Error              6.079 (df = 244)       5.202 (df = 208)
## F Statistic                   38.519*** (df = 3; 244) 25.983*** (df = 8; 208)
## ===========================================================================
## Note:                                            *p<0.1; **p<0.05; ***p<0.01
```

```r
stargazer(rdd_model_with_propagated_data,title= ("Complex Model Regression Results"), type = "text")
```

```
##
## Complex Model Regression Results
## ==========================================================
##                                   Dependent variable:
##                               ---------------------------
##                                          Credit
## ----------------------------------------------------------
## Year_centered                          -0.634***
##                                         (0.093)
##
## Post1994                               -4.453***
##                                         (1.438)
##
## Year_centered_Post1994                 1.369***
##                                         (0.115)
##
## disbursements_total                     0.029**
##                                         (0.013)
##
## customs_tariff_rate                    -0.389**
##                                         (0.163)
##
## vat_rate                                -0.466
##                                         (0.309)
##
```

```
## nontariff_barriers_eliminated          -4.435**
##                                         (1.790)
##
## total_taxation                          -0.152*
##                                         (0.078)
##
## Constant                               32.779***
##                                         (3.595)
##
## ---------------------------------------------------------
## Observations                              217
## R2                                       0.500
## Adjusted R2                              0.481
## Residual Std. Error                5.202 (df = 208)
## F Statistic                      25.983*** (df = 8; 208)
## =========================================================
## Note:                          *p<0.1; **p<0.05; ***p<0.01
```

Models comparison:

Discussion of Findings:

```
1.  Year_centered:
•   Both models show a significant negative coefficient for Year_centered, indicating a downward trend i
2.  Post1994:
•   In the simple model, the coefficient for Post1994 is not statistically significant, suggesting no in
3.  Year_centered_Post1994:
•   Both models show a significant positive coefficient for Year_centered_Post1994, indicating an upward
4.  Additional Variables in Complex Model:
•   disbursements_total: Positive and significant, indicating that higher disbursements are associated w
•   customs_tariff_rate: Negative and significant, suggesting that higher tariff rates are associated wi
•   vat_rate: Negative but not significant, indicating no clear impact on credit availability.
•   nontariff_barriers_eliminated: Negative and significant, suggesting that eliminating non-tariff bari
•   total_taxation: Negative and marginally significant, indicating that higher total taxation is associ
5.  Model Fit:
•   The complex model has a higher R-squared (0.500) and Adjusted R-squared (0.481) compared to the simp
•   The residual standard error is lower in the complex model (5.202) compared to the simple model (6.07
```

Conclusion:

The comparison between the simple and complex models highlights the importance of including additional variables to control for other factors affecting credit availability. The complex model provides a more nuanced and accurate understanding of the impact of the 1994 devaluation, revealing a significant immediate negative impact that was not apparent in the simple model. The additional variables in the complex model further elucidate the factors influencing credit availability, leading to a better overall fit and more precise estimates.

Summary of the Work Done So Far

The data analysis began with the collection of domestic credit data for WAEMU and non-WAEMU countries from 1985 to 2015. Missing values in the dataset were addressed using mean imputation. The data was then categorized and visualized to compare trends in domestic credit availability between WAEMU and non-WAEMU countries. A Regression Discontinuity Design (RDD) was employed to assess the impact of the 1994 devaluation, with the model initially showing issues with multicollinearity, which was addressed by centering the year variable. The RDD analysis included additional economic variables such as disbursements, customs tariff rates, VAT rates, and non-tariff barriers. The final analysis provided insights into the immediate

and long-term effects of the devaluation on domestic credit availability, highlighting the significant impacts of various economic reforms and interventions.

8. Difference-in-Differences (DiD) Approach:

The Difference-in-Differences (DiD) approach compares changes in outcomes over time between a treatment group (WAEMU countries) and a control group (non-WAEMU countries). This method helps to isolate the effect of the 1994 devaluation by accounting for trends that would have occurred in the absence of the devaluation.

- Causal Inference:

DiD is a robust quasi-experimental design that allows for causal inference by comparing the changes in outcomes over time between a treatment group (WAEMU countries) and a control group (non-WAEMU countries). This helps in isolating the effect of the 1994 devaluation and other policy changes on credit access from other confounding factors.

- Policy evaluation:

By comparing WAEMU countries, which experienced the devaluation, to non-WAEMU countries that did not, we can evaluate the specific impact of the devaluation and associated policies. This helps in understanding the effectiveness of these policies in improving credit access.

```r
# Load necessary libraries
library(dplyr)
library(tidyr)
library(ggplot2)
library(lmtest)
library(sandwich)

# List of WAEMU countries excluding Guinea-Bissau
waemu_countries <- c("Benin", "Burkina Faso", "Cote d'Ivoire", "Mali", "Niger", "Senegal", "Togo")

# Filter the dataset for WAEMU countries
waemu_subset <- domestic_credit_subset %>%
  filter(`Country Name` %in% waemu_countries)

# Convert data to long format
domestic_credit_long <- domestic_credit_subset %>%
  pivot_longer(cols = `1985`:`2015`, names_to = "Year", values_to = "Credit") %>%
  mutate(
    Year = as.numeric(Year),
    WAEMU = ifelse(`Country Name` %in% waemu_countries, 1, 0),
    Post1994 = ifelse(Year >= 1994, 1, 0),
    WAEMU_Post1994 = WAEMU * Post1994
  )

# Fit the simple DiD model
did_model_simple <- lm(Credit ~ WAEMU + Post1994 + WAEMU_Post1994, data = domestic_credit_long)

# Summarize the model
summary(did_model_simple)
```

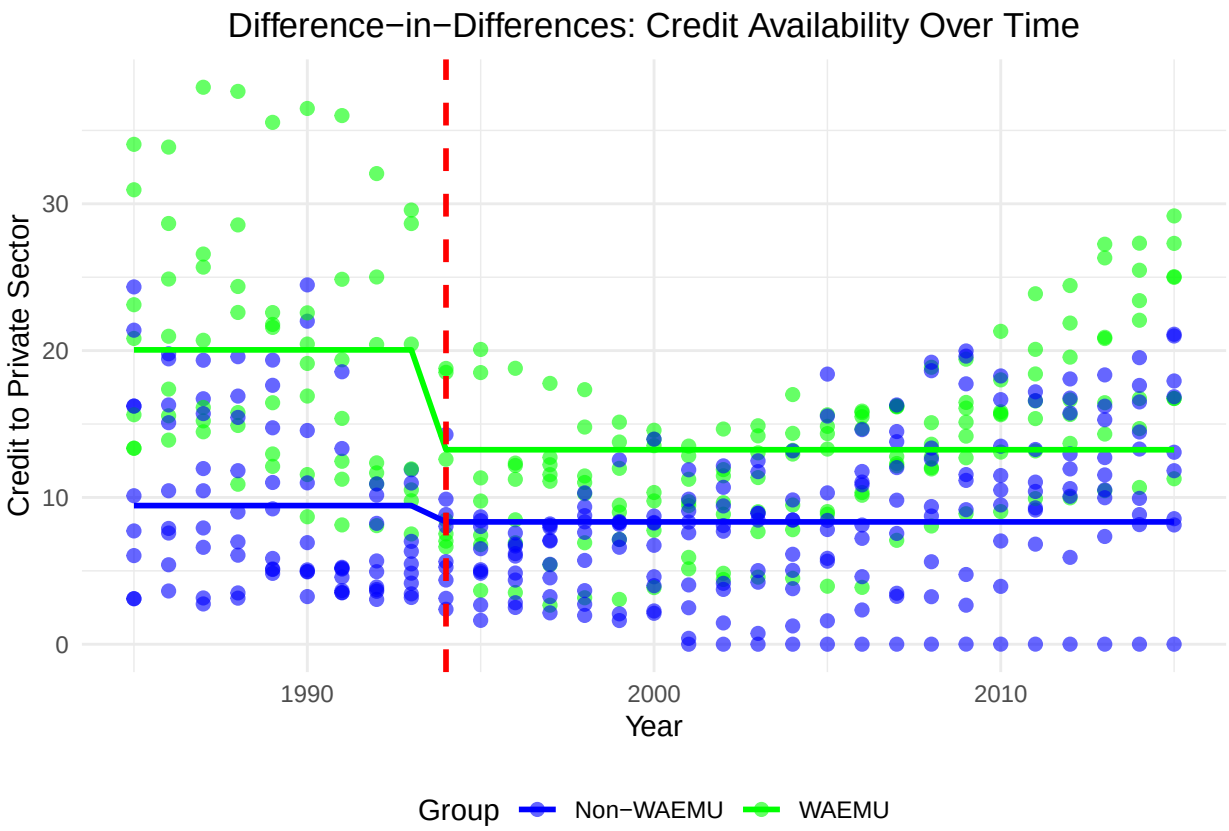
```
##
## Call:
## lm(formula = Credit ~ WAEMU + Post1994 + WAEMU_Post1994, data = domestic_credit_long)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5398  -4.3900  -0.2959   3.3432  17.8874
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)      9.4461     0.6613  14.283  < 2e-16 ***
## WAEMU           10.6057     0.9999  10.607  < 2e-16 ***
## Post1994        -1.1117     0.7851  -1.416    0.157
## WAEMU_Post1994  -5.6919     1.1869  -4.796 2.15e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.952 on 492 degrees of freedom
## Multiple R-squared:  0.2981, Adjusted R-squared:  0.2938
## F-statistic: 69.66 on 3 and 492 DF,  p-value: < 2.2e-16
```

```r
# Robust standard errors
coeftest(did_model_simple, vcov = vcovHC(did_model_simple, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                Estimate Std. Error t value  Pr(>|t|)
## (Intercept)     9.44605    0.66598 14.1837 < 2.2e-16 ***
## WAEMU          10.60566    1.24365  8.5278 < 2.2e-16 ***
## Post1994       -1.11170    0.76248 -1.4580    0.1455
## WAEMU_Post1994 -5.69189    1.37487 -4.1399 4.088e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Generate predicted values for visualization
domestic_credit_long$Predicted_Credit <- predict(did_model_simple, newdata = domestic_credit_long)

# Create the plot
ggplot(domestic_credit_long, aes(x = Year, y = Credit, color = factor(WAEMU))) +
  geom_point(size = 2, alpha = 0.6) +
  geom_line(aes(y = Predicted_Credit), size = 1) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "red", size = 1) +
  labs(title = "Difference-in-Differences: Credit Availability Over Time",
       x = "Year",
       y = "Credit to Private Sector",
       color = "Group") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        legend.position = "bottom") +
  scale_color_manual(values = c("0" = "blue", "1" = "green"), labels = c("Non-WAEMU", "WAEMU"))
```




```r
stargazer(did_model_simple,title= ("DiD Regression Results"), type = "text")
```

```
##
## DiD Regression Results
## ===============================================
##                         Dependent variable:
##                     ---------------------------
##                              Credit
## -----------------------------------------------
## WAEMU                      10.606***
##                             (1.000)
##
## Post1994                    -1.112
##                             (0.785)
##
## WAEMU_Post1994             -5.692***
##                             (1.187)
##
## Constant                   9.446***
##                             (0.661)
##
## -----------------------------------------------
## Observations                  496
## R2                           0.298
## Adjusted R2                  0.294
## Residual Std. Error      5.952 (df = 492)
## F Statistic          69.657*** (df = 3; 492)
## ===============================================
```

```
## Note:                *p<0.1; **p<0.05; ***p<0.01
```

Interpretation of Coefficients

```
1.  Intercept (Estimate: 9.4461, p-value: < 2e-16)
•   This represents the baseline level of credit availability to the private sector in non-WAEMU countri
2.  WAEMU (Estimate: 10.6057, p-value: < 2e-16)
•   This coefficient represents the difference in credit availability between WAEMU and non-WAEMU counti
3.  Post1994 (Estimate: -1.1117, p-value: 0.157)
•   This coefficient represents the change in credit availability in non-WAEMU countries after 1994. The
4.  WAEMU_Post1994 (Estimate: -5.6919, p-value: 2.15e-06)
•   This interaction term represents the additional effect of being a WAEMU country after 1994. The esti
```

Statistical Metrics

```
•   Residual standard error: 5.952
•   This value provides a measure of the average distance that the observed values fall from the regress
•   Multiple R-squared: 0.2981
•   This value indicates that approximately 29.81% of the variance in credit availability is explained b
•   Adjusted R-squared: 0.2938
•   The adjusted R-squared takes into account the number of predictors in the model and provides a more
•   F-statistic: 69.66 on 3 and 492 DF, p-value: < 2.2e-16
•   The F-statistic tests the overall significance of the model. A p-value < 2.2e-16 indicates that the
```

Discussion of Relevance

The results of your DiD analysis indicate that, while there was no significant change in credit availability in non-WAEMU countries after 1994 (Post1994 coefficient), WAEMU countries experienced a significant additional reduction in credit availability (WAEMU_Post1994 coefficient). This suggests that the 1994 devaluation had a substantial negative impact on credit availability in WAEMU countries compared to non-WAEMU countries.

```r
# Fit the simple DiD model
did_model_simple <- lm(Credit ~ WAEMU + Post1994 + WAEMU_Post1994, data = domestic_credit_long)

# Summarize the model
summary(did_model_simple)
```

```
##
## Call:
## lm(formula = Credit ~ WAEMU + Post1994 + WAEMU_Post1994, data = domestic_credit_long)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5398  -4.3900  -0.2959   3.3432  17.8874
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)      9.4461     0.6613  14.283  < 2e-16 ***
## WAEMU           10.6057     0.9999  10.607  < 2e-16 ***
## Post1994        -1.1117     0.7851  -1.416    0.157
## WAEMU_Post1994  -5.6919     1.1869  -4.796 2.15e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.952 on 492 degrees of freedom
```

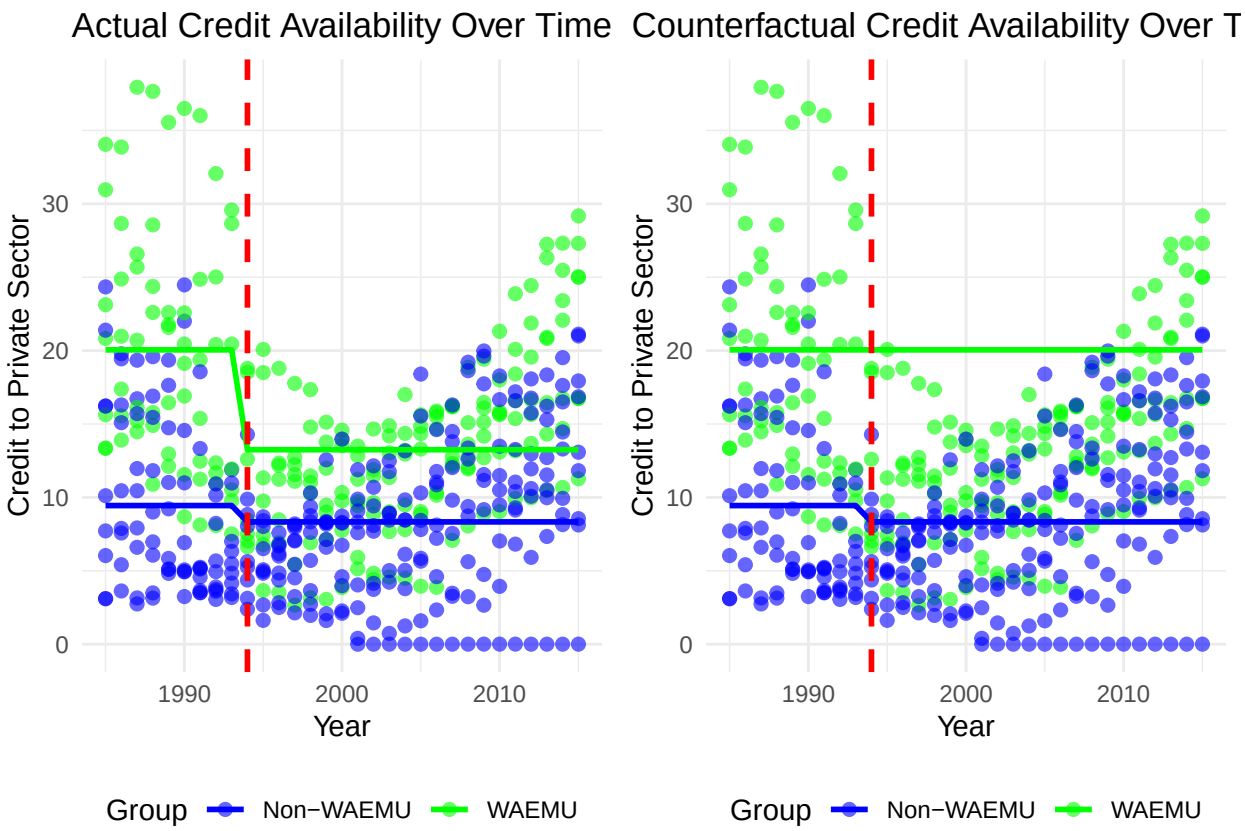
```
## Multiple R-squared:  0.2981, Adjusted R-squared:  0.2938
## F-statistic: 69.66 on 3 and 492 DF,  p-value: < 2.2e-16
```

```r
# Create a counterfactual scenario where Post1994 is set to 0 for WAEMU countries
domestic_credit_long_counterfactual <- domestic_credit_long %>%
  mutate(Post1994 = ifelse(WAEMU == 1, 0, Post1994),
         WAEMU_Post1994 = WAEMU * Post1994)

# Predict credit values under the counterfactual scenario
domestic_credit_long_counterfactual$Counterfactual_Credit <- predict(did_model_simple, newdata = domesti

# Predict actual values with the simple model for comparison
domestic_credit_long$Predicted_Credit <- predict(did_model_simple, newdata = domestic_credit_long)
```

```r
# Plot the actual data
actual_plot <- ggplot(domestic_credit_long, aes(x = Year, y = Credit, color = factor(WAEMU))) +
  geom_point(size = 2, alpha = 0.6) +
  geom_line(aes(y = Predicted_Credit), size = 1) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "red", size = 1) +
  labs(title = "Actual Credit Availability Over Time",
       x = "Year",
       y = "Credit to Private Sector",
       color = "Group") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        legend.position = "bottom") +
  scale_color_manual(values = c("0" = "blue", "1" = "green"), labels = c("Non-WAEMU", "WAEMU"))

# Plot the counterfactual data
counterfactual_plot <- ggplot(domestic_credit_long_counterfactual, aes(x = Year, y = Credit, color = fa
  geom_point(size = 2, alpha = 0.6) +
  geom_line(aes(y = Counterfactual_Credit), size = 1) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "red", size = 1) +
  labs(title = "Counterfactual Credit Availability Over Time",
       x = "Year",
       y = "Credit to Private Sector",
       color = "Group") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        legend.position = "bottom") +
  scale_color_manual(values = c("0" = "blue", "1" = "green"), labels = c("Non-WAEMU", "WAEMU"))

# Print both plots side by side
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'

## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
grid.arrange(actual_plot, counterfactual_plot, ncol = 2)
```




```
# Generate predicted values for visualization
domestic_credit_long$Predicted_Credit <- predict(did_model_simple, newdata = domestic_credit_long)

# Create a counterfactual scenario where Post1994 is set to 0 for WAEMU countries
domestic_credit_long_counterfactual <- domestic_credit_long %>%
  mutate(Post1994 = ifelse(WAEMU == 1, 0, Post1994),
         WAEMU_Post1994 = WAEMU * Post1994)

# Predict credit values under the counterfactual scenario
domestic_credit_long_counterfactual$Counterfactual_Credit <- predict(did_model_simple, newdata = domesti

# Aggregate data for smooth line plotting
actual_data <- domestic_credit_long %>%
  group_by(Year, WAEMU) %>%
  summarize(Predicted_Credit = mean(Predicted_Credit, na.rm = TRUE))
```

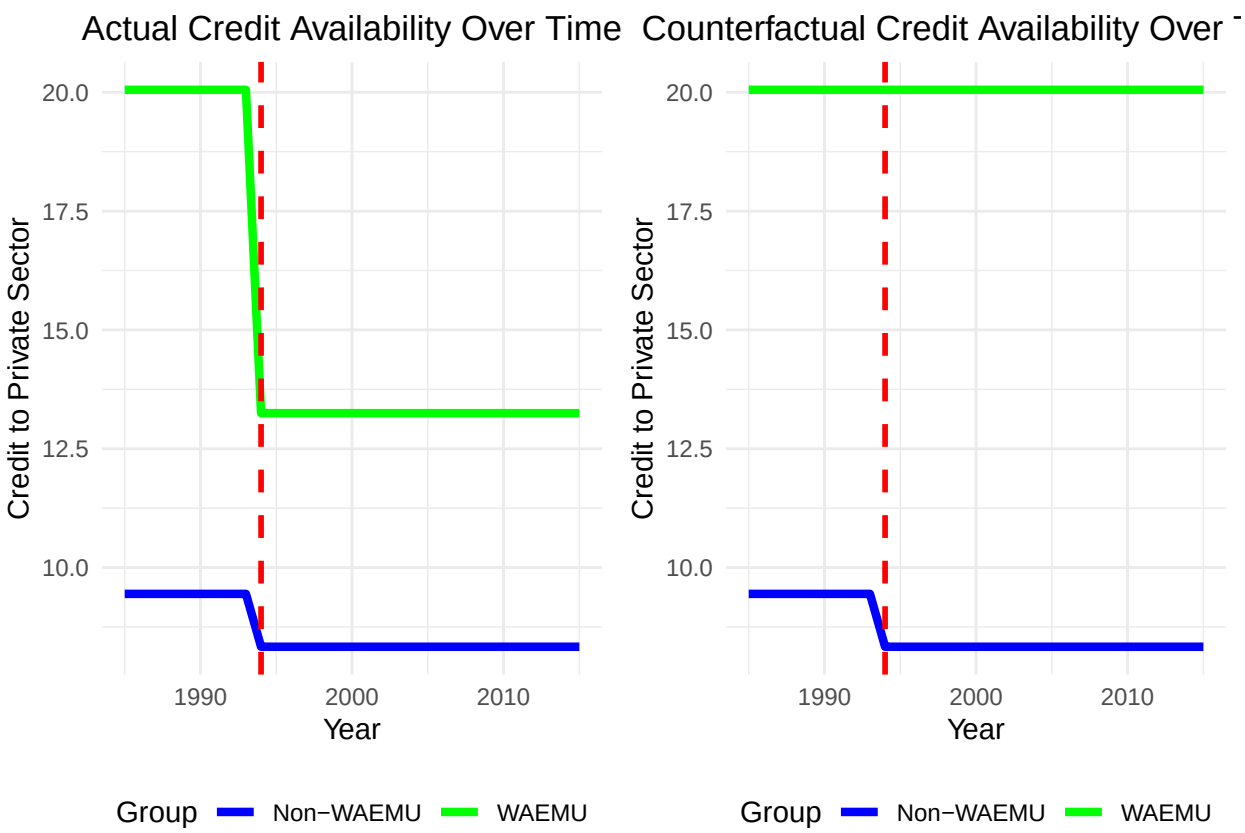
```
## `summarise()` has grouped output by 'Year'. You can override using the
## `.groups` argument.
```

```
counterfactual_data <- domestic_credit_long_counterfactual %>%
  group_by(Year, WAEMU) %>%
  summarize(Counterfactual_Credit = mean(Counterfactual_Credit, na.rm = TRUE))
```

```
## `summarise()` has grouped output by 'Year'. You can override using the
## `.groups` argument.
```

```
# Plot the actual data
actual_plot <- ggplot(actual_data, aes(x = Year, y = Predicted_Credit, color = factor(WAEMU))) +
  geom_line(size = 1.5) +
```

```r
  geom_vline(xintercept = 1994, linetype = "dashed", color = "red", size = 1) +
  labs(title = "Actual Credit Availability Over Time",
       x = "Year",
       y = "Credit to Private Sector",
       color = "Group") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        legend.position = "bottom") +
  scale_color_manual(values = c("0" = "blue", "1" = "green"), labels = c("Non-WAEMU", "WAEMU"))

# Plot the counterfactual data
counterfactual_plot <- ggplot(counterfactual_data, aes(x = Year, y = Counterfactual_Credit, color = fac
  geom_line(size = 1.5) +
  geom_vline(xintercept = 1994, linetype = "dashed", color = "red", size = 1) +
  labs(title = "Counterfactual Credit Availability Over Time",
       x = "Year",
       y = "Credit to Private Sector",
       color = "Group") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5),
        legend.position = "bottom") +
  scale_color_manual(values = c("0" = "blue", "1" = "green"), labels = c("Non-WAEMU", "WAEMU"))

# Print both plots side by side
library(gridExtra)
grid.arrange(actual_plot, counterfactual_plot, ncol = 2)
```



The graphs effectively illustrate the significant negative impact of the 1994 devaluation on credit availability in WAEMU countries. The counterfactual analysis provides a compelling visualization of what credit availability might have looked like in the absence of the devaluation, reinforcing the conclusion that the devaluation played a crucial role in reducing credit levels in WAEMU countries.

Analysis and Commentary on the Graphs

Actual Credit Availability Over Time

The left plot titled "Actual Credit Availability Over Time" shows the trend of credit availability to the private sector in WAEMU and Non-WAEMU countries over time, with a focus on the period before and after the 1994 devaluation.

```
1.  WAEMU Countries (Green Line):
•   The credit availability is relatively stable before 1994.
•   After the 1994 devaluation (indicated by the red dashed line), there is a noticeable drop in credit
•   The credit availability remains lower after the devaluation and does not show signs of recovery witl
2.  Non-WAEMU Countries (Blue Line):
•   The credit availability is relatively stable before 1994.
•   There is no significant change in credit availability around the 1994 period.
•   The credit availability remains stable after 1994, indicating that these countries were not affecte
```

Counterfactual Credit Availability Over Time

The right plot titled "Counterfactual Credit Availability Over Time" shows a hypothetical scenario where the 1994 devaluation did not occur.

```
1.  WAEMU Countries (Green Line):
•   The credit availability remains stable throughout the period, including after 1994.
•   This suggests that without the devaluation, WAEMU countries might have maintained their pre-1994 lev
2.  Non-WAEMU Countries (Blue Line):
•   The trend is identical to the actual scenario, indicating stability in credit availability over time
```

Interpretation

```
•   Impact of the 1994 Devaluation on WAEMU Countries:
•   The actual data shows a significant negative impact on credit availability in WAEMU countries follo
•   The counterfactual scenario supports the idea that the devaluation was a major factor in reducing c
•   Non-WAEMU Countries:
•   These countries serve as a control group, showing no significant changes around the 1994 period, wh
```

Conclusion

The graphs effectively illustrate the significant negative impact of the 1994 devaluation on credit availability in WAEMU countries. The counterfactual analysis provides a compelling visualization of what credit availability might have looked like in the absence of the devaluation, reinforcing the conclusion that the devaluation played a crucial role in reducing credit levels in WAEMU countries.